AI for Talent Identification and Developing Rural Sports in Sri Lanka

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Introduction: Sri Lanka's sports sector struggles with unequal talent discovery, particularly in rural regions where access to structured training and competitive exposure is limited. Many young athletes with potential remain unnoticed due to lack of opportunity, resources, and data-driven selection systems. To address this disparity, a technological intervention is required to ensure fair and efficient talent identification across the country.

Objectives: This study aims to design and develop AthleteConnect, a web-based AI-powered platform to identify, guide, and promote young athletic talent from underserved rural areas, while providing a structured, data-driven pathway from discovery to professional development.

Methodology: AthleteConnect uses an intelligent matching system that analyzes age, body metrics, and medical history to suggest suitable sports. Matched athletes enter a personalized training program. Role-based dashboards allow athletes, coaches, and officials to manage profiles, upload videos, and track progress. A TensorFlow Lite and OpenCV-powered scouting engine analyzes videos and stats to generate talent scores. The platform uses Firebase Auth, PostgreSQL, and Google Cloud Storage, with Firebase real-time messaging.

Results & Discussion: At this stage, the prototype is not yet implemented, but its design supports role-based registration, video uploads, AI-driven scoring, and Google Maps-based athlete mapping. A pilot rollout involving 50 athletes and 10 coaches across three rural districts is planned. The project anticipates achieving ≥80% athlete–coach adoption during the pilot and an AI accuracy rate of over 80% in talent score generation. These projected outcomes aim to improve fairness in athlete selection, increase rural visibility, and provide policymakers with actionable insights into regional sports development.

Conclusion: Although development and real-world testing are yet to begin, AthleteConnect represents a scalable, inclusive solution to bridge the rural–urban sports divide in Sri Lanka. By combining requirements-driven design with AI-supported talent analysis, the project lays a strong foundation for empowering rural youth, and modernizing athlete identification. In future will focus on prototype implementation, testing, and national level adoption.

Keywords: Talent Identification, AI-Based Sport Recommendation, Rural Sports Development and Video Performance Analysis, Regional Mapping, Role-Based Access and Real-Time Coach Interaction